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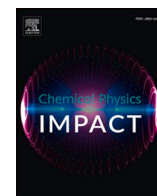


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Design, 3D-QSAR, molecular docking, ADMET, molecular dynamics and MM-PBSA simulations for new anti-breast cancer agents

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ABSTRACT

Breast cancer is the most frequent form of malignant tumor in women, and represents a major public health problem due to its high mortality rate. Although a multitude of therapeutic options exist for control of this disease, the emergence of resistance to current pharmaceutical treatments underscores the urgency of developing new anti-breast cancer drugs, with a focus on reducing the adverse effects associated with current therapeutic agents. The present study concerns a new series of (23) compounds based on 1,4-quinone and quinoline derivatives to design candidate drugs against breast cancer. For this purpose, integrated computational techniques were applied, including 3D-QSAR, molecular docking and molecular dynamics simulations (MD). CoMFA and CoMSIA were used to build a robust and highly reliable 3D-QSAR models. To validate the model's predictive capabilities, an external validation was carried out. The results of the best model (CoMSIA/SEA) revealed that electrostatic, steric and hydrogen bond acceptor fields had a significant effect on the anti-breast cancer activity of molecules studied. In addition, evaluation of ADMET properties determined whether these newly designed ligands were likely to be selected as drug-candidates. To confirm the binding stability of the selected ligands to aromatase (3S7S) and validate the molecular docking results, molecular dynamics simulations lasting 100 nanoseconds were performed by calculating RMSD, RMSF, RoG, H-bond, SASA and MM-PBSA parameters. As a result, only one designed compound (ligand 5) emerged as the most promising drug candidate for experimental in vitro and in vivo testing, due to its potential inhibition of breast cancer.

1. Introduction

Based on WHO (World health organization) data, breast cancer is the second most prevalent cause of death worldwide [1,2]. Despite the many efforts and therapeutic approaches developed over the years to treat this cancer including surgery, radiotherapy and chemotherapy, their effectiveness in terms of improving survival rates remains limited [3]. Biochemotherapy is based on the use of several drugs that target cancer-causing proteins and enzymes in order to inhibit or completely stop their action [4]. Accordingly, different types of enzymes that play an important role in cancer development have been targeted including estrogen receptors (ER) and aromatase enzymes that play a prominent role in the development of breast cancer linked to hormones [5]. Therefore, aromatase has been chosen as a potential therapeutic target

for the treatment of breast cancer, stimulating the conversion of androstenedione into estrogen [6]. Exemestane has emerged as one of the most effective aromatase-targeted drugs for the treatment of breast cancer, but it has been confronted with problems of resistance and adverse effects [7,8]. Researchers have explored various molecular structures to formulate inhibitors that can effectively target breast cancer. However, the complex and heterogeneous nature of this disease limits the effectiveness of conventional targeted therapies, leading to an increase in side-effects and the emergence of multi-drug resistance. For this, the discovery of new molecules combining two or more active pharmacophores has proved to be a promising approach to the development and synthesis of anti-breast cancer drugs. Thus, Quinone and quinoline compounds have been studied as aromatase inhibitors for their potential pharmacological activity against breast cancer [9].

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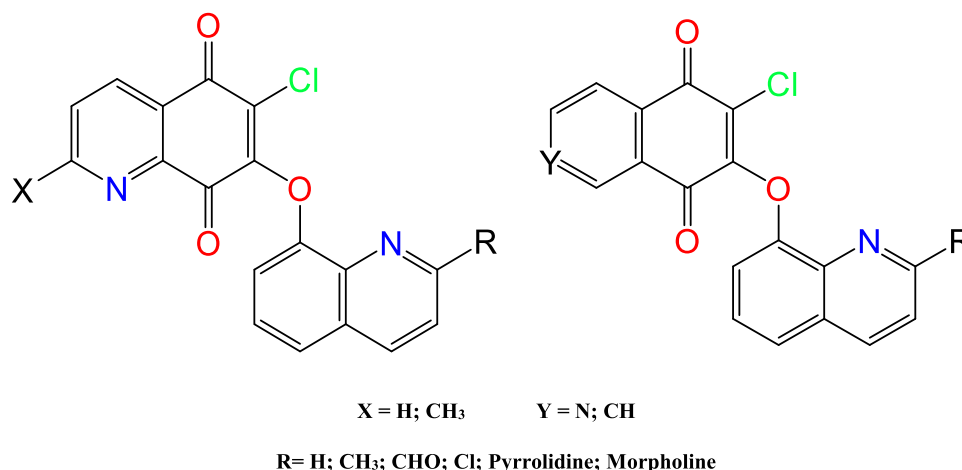


Fig. 1. Structures of the investigated Quinone and Quinoline derivatives.

Moreover, these compounds have also been identified as a new family of bioactive chemicals as potential inhibitors of the proliferation of other types of cancer such as cervical cancer, melanoma, hepatoma, colorectal cancer, and lung cancer, in addition to their antifungal and anti-inflammatory properties [10–12].

For more cost-effective development of medicines, the use of computational methods, particularly in silico techniques or computer-aided drug design (CADD) coupled with structure-based drug design (SBDD) and ligand-based drug design (LBDD), plays an essential role in the discovery of new drug-candidates. Thus, rational integrated methods such as QSAR (quantitative structure-activity relationship) [13] and ADMET (absorption, distribution, metabolism, excretion and toxicity) have emerged as effective strategies for the discovery and design of potential new drugs [14,15]. Molecular docking and molecular dynamics simulations were also used to find the crucial amino acids involved in the interactions between the active site of enzymes and the ligands studied [16]. In fact, 3D-QSAR models, by creating comparative molecular field analysis (CoMFA) and Comparative Molecular Similarity Index Analysis (CoMSIA), may be used to study the effects of relevant descriptors (fields) on the efficacy of treatment of the disease under investigation as well as to discover suitable drug-candidates [17]. For this, each model then needed to be validated internally and externally, followed by a simulation of the molecular docking of the newly designed compounds using a suitable software such as AutoDock Vina [18]. In addition, the pharmacokinetics of drug-candidates can be explained using ADMET properties [19]. While Molecular dynamics simulations (MD) may confirm the docking results, aiming to analyze protein-ligand complexes' conformational stability and fluctuations [20]. Alternatively, the binding free energy estimation confirms the binding affinity and the resulting trajectory can extract the pertinent metrics like the root mean square deviation (RMSD), root mean square fluctuations (RMSF), solvent accessible surface area (SASA), the radius of gyration (RoG), changes in hydrogen bond count, and the separation between the center of masses of the ligand and aromatase protein [21–28].

In this research contribution, the test for the treatment of breast cancer with 1,4-quinone and quinoline derivatives was investigated in silico in order to discover potential new drug-candidates. For this purpose, 3D-QSAR and molecular docking techniques were used to select the pertinent molecular descriptors influencing the anti-breast cancer activity as well as the types of interactions with the active site of aromatase. In addition, ADMET predictions have been used to assess specific pharmacokinetic properties as well as to virtually screen designed molecules against breast cancer. Finally, MD parameters including RMSD, RMSF and PCA (principal component analysis) as well as binding free energy, were used to validate the molecular docking results and the stability study of selected drug-candidates, via virtual screening

Table 1

Anti-breast cancer activities of the 23 Quinone and Quinoline derivatives.

N°	IC ₅₀ (μM)	pIC ₅₀	X	Y	R
1	19.350	4.713	H	–	H
2	2.110	5.676	H	–	CH ₃
3	18.330	4.737	H	–	CHO
4	2.110	5.676	H	–	Cl
5	02.030	5.693	H	–	Pyrrolidine
6	16.210	4.790	H	–	Morpholine
7	02.090	5.680	CH ₃	–	H
8	01.780	5.750	CH ₃	–	CH ₃
9	19.110	4.719	CH ₃	–	CHO
10	17.410	4.759	CH ₃	–	Cl
11	01.890	5.724	CH ₃	–	Pyrrolidine
12	14.110	4.850	CH ₃	–	Morpholine
13	01.160	5.936	–	N	H
14	00.610	6.215	–	N	CH ₃
15	00.890	6.051	–	N	CHO
16	01.750	5.757	–	N	Cl
17	16.850	4.773	–	N	Pyrrolidine
18	01.030	5.987	–	N	Morpholine
19	14.070	4.852	–	CH	H
20	16.310	4.788	–	CH	CH ₃
21	11.660	4.933	–	CH	Cl
22	24.050	4.619	–	CH	Pyrrolidine
23	17.450	4.758	–	CH	Morpholine

$$pIC_{50} = -\log IC_{50}.$$

technique, for in vitro and in vivo testing.

2. Material and methods

2.1. Breast cancer data sets

To build 3D-QSAR models for a set of twenty-three 1,4-quinone and quinoline derivatives (As shown in Fig. 1 and Table 1), the dataset was randomly divided into two distinct subsets [9]. The training set was used to develop the model, while the test set was utilized to validate it.

2.2. QSAR study

2.2.1. Molecular alignment and CoMFA/CoMSIA models generation

A crucial step in the development of the CoMFA and CoMSIA models using partial least squares (PLS) regression is the alignment of the molecules [29]. To achieve this, an initial sketch of molecular structures was made using the SYBYL-X.2.1 platform [30] which was followed by their optimisation via the Tripos force field and Gasteiger-Hückel atomic partial charges. By applying Powell's gradient approach, a considerable

number of 100,000 iterations were required to obtain a structurally stable conformation [31].

Steric (S) and electrostatic (E) fields, generated by the sp^3 hybridization of a carbon atom with a Van Der Waals radius of 1.52 and a net charge of 1.0, have been applied to the CoMFA model. In contrast, the CoMSIA model is based on the use of a probe atom to obtain more detailed physicochemical descriptors, including hydrophobic (H), hydrogen bond donor (D) and acceptor (A) fields, as well as steric and electrostatic fields. This was achieved by setting initial attenuation factors at 0.3 kcal/mol and column filtering at 2.0 kcal/mol [32]. Next, applying the partial least squares (PLS) regression method and considering the results relating to the number of principal components (N), several models can be developed to explore the correlation between the CoMFA and CoMSIA fields and anti-breast cancer activities [33].

2.2.2. Statistical analysis and validation of 3D-QSAR models

To evaluate the global significance of the models, the highest value of coefficient of determination R^2 (Eq. (1)), the lowest value of mean squared error MSE (Eq. (2)), the Fisher's statistical parameter (F-value) and the level of significance (p-value) are the primary parameters used in statistical analysis of QSAR models as well as the highest coefficient of determination of cross-validation (Q^2) that must be more than 0.5 according to Eq. (3) [34]:

$$R^2 = 1 - \frac{\sum_{i=1}^n (Y_{iobs} - Y_{ical})^2}{\sum_{i=1}^n (Y_{iobs} - \bar{Y}_{cal})^2} \quad (1)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_{iobs} - Y_{ical})^2 \quad (2)$$

$$Q^2 = 1 - \frac{\sum_{j=1}^N (Y_{jobs}(train) - Y_{jcal}(train))^2}{\sum_{j=1}^N (Y_{jobs}(train) - \bar{Y}_{cal}(train))^2} \quad (3)$$

Where: Y_{iobs} represents the value of the observed response i , Y_{ical} is the value of the predicted response i , $\bar{Y}_{cal}$ represents the average value of predicted responses, $Y_{jobs}^2(train)$ is the observed response's value, $Y_{jcal}(train)$ is the value predicted by cross validation and $\bar{Y}_{cal}(train)$ is the mean of predicted response values.

To assess the predictive power of the 3D-QSAR models, the anti-breast cancer activities of the molecules included in the test set were re-estimated. To do this, the external coefficient of determination R_{pred}^2 , which must be greater than 0.6, was used as a measure to assess the reliability of the models generated [35]. To confirm the reliability of the QSAR model, a Y-randomization test was carried out. In this procedure, Y (Response values) were randomly permitted into the X matrix (descriptor matrix) [36]. Accordingly, the correlation coefficient of the randomized model (R_{rand}^2) must be less than the R^2 of the non-randomized model and a supplementary coefficient CR_p^2 (Eq. (4)) must be greater than 0.5 [29,34]:

$$cR_p^2 = R^2 \times \sqrt{(R^2 - R_{rand}^2)} \quad (4)$$

In addition, to check the validity of the model and whether it is the result of a random correlation, an evaluation approach based on the values of Q_{rand}^2 (coefficient of determination of cross-validation for the Y randomisation test) was introduced using the following four specific inequalities:

- $Q_{rand}^2 < 0.2$: No possibility of a relationship by chance.
- $0.2 < Q_{rand}^2 < 0.3$: Negligible probability of a relationship by chance
- $0.3 < Q_{rand}^2 < 0.4$: Not a significant random relationship
- $Q_{rand}^2 < 0.4$: Seem to be correlated by chance

The applicability domain (AD) of the best model was then determined using the leverage technique (William diagram) to specify the region of chemical space where predictions of anti-breast cancer activity are reliable. To achieve this, the most usual method is to evaluate the leverage value h_i for each individual compound, as indicated in Eq. (5). Accordingly, a compound is considered to be outside the applicability domain if its leverage h_i exceeds the threshold h^* as defined in Eq. (6) [37]:

$$h_i = x_i^T (X^T X)^{-1} x_i \quad (5)$$

$$h^* = 3 \times \frac{(K + 1)}{n} \quad (6)$$

where i varies between 1 and n , x_i refers to a vector characterizing the chemical compound to be identified, and X represents a matrix containing the k values of the pertinent descriptors model corresponding to n compounds in the training set.

2.2.3. Molecular docking modeling

For obtaining the kinds of interactions between the chosen ligands and the targeted active site responsible for breast cancer inhibition, molecular docking modelling has been applied. Specifically, the most active molecule and the least active molecule, as well as the designed drug-candidates, were docked into the aromatase active site (PDB: 3S7S) [38]. First, the receptor was purified using Discovery Studio software, removing all original ligands, water molecules and other non-protein components. Next, the ligand-protein interactions were analysed using Auto-Dock version 1.5.6 [18]. Accordingly, AUTOGUID was used to calculate the grid's dimensions, and Discovery Studio was also used to explore examine the created interactions in the two- and three-dimensional representations. The co-crystallized ligand was re-docked in confirm the validity of the docking process, and the root mean square deviation (RMSD), which needs to be less than two angstroms, was calculated [39].

2.2.4. Pharmacokinetic and drug likeness investigations

Effective and useful drug molecules are defined by their pharmacodynamic and pharmacokinetic properties, including significant parameters like high therapeutic potential, affinity, acceptable levels of toxicity, as well as selectivity for the protein binding target [40,41]. This investigation included the use of online tools like pkCSM [42] and SwissADME [43] to examine drug similarity features and chemical absorption, distribution, metabolism, excretion, and toxicity (ADMET). To do this, in silico predictions were integrated about intestinal absorption, blood-brain barrier and penetration into the central nervous system, biotransformation, clearance, and the AMES test of potential drug-candidates [41,43,44].

2.3. Molecular dynamic simulations (MD)

Molecular Dynamics (MD) calculations were performed on the best docking poses [45] to ascertain the stability of the interaction between the protein and ligand. The input files for the MD calculations were generated using the CHARMM-GUI solution builder, which employed the CHARMM force field parameters for proteins. The topology of the ligands was generated via the Param-Chem server using the CHARMM General Forcefield. The CHARMM-GUI solution builder comprises five steps. In the first step, the tool reads the coordinates of the protein-ligand complex. The second step involves the solvation of the protein-ligand complex, determining the system's shape and size. Na^+ and Cl^- ions are added in this step to neutralize the system. In the third step, periodic boundary conditions (PBC) are set to approximate an extensive system using a unit cell replicated in all directions. The simulation runs only for atoms within the PBC box, and erroneous contacts are removed by running a short minimization. The fourth and

fifth steps involve system equilibration and production. Equilibration is done in NVT and NPT ensembles to ensure the system achieves the desired temperature and pressure. The input files for equilibration and production are downloaded, and desired changes are made, including several steps of MD run, frequency of saving of trajectories, calculation of energy, etc. All MD calculations, equilibration, and production runs were performed using GROMACS 2020.2. Initially, all complexes were solvated in a cubic box of TIP3P waters. Na⁺ and Cl⁻ ions were added to neutralize the net atomic charge of the entire system by randomly replacing water molecules. The PBC was imposed, considering the system's shape and size. Non-bonded interactions were treated with a 12 Å cutoff distance. The neighbor searching list was buffered with the Verlet cutoff scheme, and long-range electrostatic interactions were treated with the particle mesh Ewald (PME) method. The protein-ligand complex was subjected to the CHARMM36 forcefield. Before the production simulation, the system underwent energy minimization using the steepest descent algorithm (5000 steps). The complex was then equilibrated by subjecting it to NVT and NPT ensembles and simulating it for 125 ps at 300.15 K using 400 kJ/mol.nm² and 40 kJ/mol. nm² positional restraints on the backbone and side chains. Finally, the complex underwent a production simulation run for 100 ns in an NPT ensemble at 300.15 K and 1 bar. The nose Hoover thermostat was used to maintain temperature; similarly, the Parrinello-Rahman barostat was used to maintain pressure. The LINCS algorithm was used to constrain H-bonds using the inputs provided by CHARMM-GUI. The V-rescale thermostat at 300 K with a coupling constant of 1 ps was used. Trajectories were stored every 2 ps. Simulations of 100 ns in NPT assembly were performed for the production stage [46,47].

2.3.1. Trajectory analysis

We used GROMACS tools to analyze molecular dynamics (MD) simulations in order to comprehend how the atoms in the protein and ligand behaved [48]. Using the very exact *gmx_rms* subprogram, we accurately computed the root mean square deviation (RMSD) between the locations of protein and ligand atoms. Additionally, we used the *gmx_rmsf* function to determine the root mean square fluctuations (RMSF) based on the protein's C-alpha atoms. Using the *gmx_gyr* tool, we also calculated the gyration radius of every protein atom. *gmx_hbond* was used to calculate the number of hydrogen bonds present at the protein-ligand interaction. Additionally, throughout the simulation, the center of the mass distance between the protein and ligand was determined using *gmx_distance*. The VMD molecular graphics application was used to examine the trajectory and protein-ligand interaction frequency, which allowed us to learn important things about how the system behaves [49].

2.3.2. Binding free energy (MM/PBSA calculation)

The systems chosen for further investigation underwent Molecular Mechanics/Poisson-Boltzmann Surface Area (MM/PBSA) computations using the GROMACS software suite's *g_mmpbsa* tool:

$$\Delta G_{\text{binding}} = \Delta G_{\text{complex}} - (\Delta G_{\text{protein}} + \Delta G_{\text{ligand}})$$

The ΔG complex represents the protein-ligand complex's overall free energy. The total free energy of the separated protein and ligand in a solvent are represented simultaneously by the ΔG protein and ΔG ligand, respectively. Each residue's energy contribution to the binding energy may be calculated using the *g_mmpbsa* tool, which can also break down the binding energy. ΔE_{MM} , ΔG_{polar} , and $\Delta G_{\text{non-polar}}$ are computed individually for each residue and then summed together to determine each residue's contribution to the binding energy. Because *g_mmpbsa* can only read files from specified GROMACS versions, GROMACS 5.1.4 recreated the binary run input file (.tpr). The molecular structure file (.gro), topology file (.top), and MD-parameter file (.mdp), which are all products of the MD process, are used to regenerate the binary run input file.

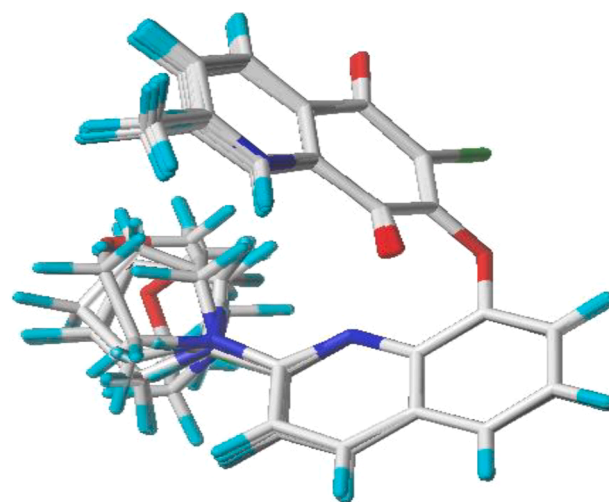


Fig. 2. The aligned compounds using molecule 14 as a template.

Table 2

Models' statistical parameters.

Generated model	Fields' description	Q ²	N	SEE	R ²	F
CoMFA / SE	Steric (S)- Electrostatic (E)	0.471	6	0.178	0.938	22.500
CoMSIA / SEH	Steric (S)- Hydrophobic (H)- Electrostatic (E)	0.417	4	0.207	0.897	23.870
CoMSIA / SED	Steric (S)- Electrostatic(E)- Donnor of hydrogen bond (D)	0.481	5	0.203	0.910	20.216
CoMSIA / SEA	Steric (S)- Electrostatic (E) - Acceptor of hydrogen bond (A)	0.527	3	0.195	0.900	36.080
CoMSIA / EHA	Electrostatic (E)- Hydrophobic (H)- Acceptor of hydrogen bond (A)	0.441	6	0.203	0.919	16.988
CoMSIA / SEHD	Steric (S)- Electrostatic (E)- Hydrophobic - Donnor of hydrogen bond (D)	0.417	4	0.207	0.897	23.870
CoMSIA / SEHA	Steric (S)- Electrostatic (E)- Hydrophobic (H) - Acceptor of hydrogen bond (A)	0.432	5	0.196	0.915	21.616
CoMSIA / SEDA	Steric (S)- Electrostatic (E)- Donnor (D) and Acceptor (A) of hydrogen bond	0.527	3	0.195	0.900	36.080
CoMSIA / EHDA	Electrostatic (E)- Hydrophobic (H)- Donnor(D) and Acceptor (A) of hydrogen bond	0.441	6	0.203	0.919	16.988
CoMSIA / SEHDA	Steric (S)- Electrostatic- Hydrophobic (H)- Donor(D) and Acceptor(A) of hydrogen bond	0.432	5	0.196	0.915	21.616

Table 3

Predicted pIC_{50} and CoMSIA descriptors for the compounds under study (*: test set).

Name	pIC_{50}	Acceptor field (A)	Steric field (S)	Electrostatic field (E)	$\text{pIC}_{50\text{pred}}$
1*	4.713	5.584	8.033	1.471	4.896
2	5.676	5.582	8.372	1.508	5.655
3*	4.737	5.063	9.623	1.604	4.706
4	5.676	5.583	8.046	1.436	5.562
5	5.693	5.579	9.627	1.608	5.696
6	4.790	6.457	9.768	1.729	5.095
7*	5.680	5.587	8.347	1.467	5.606
8	5.750	5.585	8.675	1.504	5.663
9*	4.719	5.565	8.378	1.440	4.701
10*	4.759	5.585	8.359	1.433	4.707
11	5.724	5.579	9.893	1.602	5.698
12*	4.851	6.459	10.030	1.722	5.004
13	5.936	5.019	8.054	1.456	6.060
14	6.215	5.020	8.394	1.494	6.122
15	6.051	6.509	8.294	1.628	6.066
16	5.757	5.019	8.068	1.423	6.029
17*	4.773	5.020	9.647	1.592	4.680
18	5.987	5.979	9.788	1.721	5.596
19	4.852	4.553	8.165	1.315	4.817
20	4.788	4.551	8.501	1.357	4.871
21	4.933	4.552	8.178	1.277	4.783
22	4.619	4.936	9.768	1.452	4.834
23	4.758	5.622	9.914	1.578	4.654

3. Results and discussion

3.1. Molecular structure alignment

To align the data set and provide the contour maps required for the CoMFA and CoMSIA studies, molecule 14 was chosen as the reference compound to create a reliable 3D-QSAR model as shown in Fig. 2.

3.2. 3D-QSAR models generation

In this study, 10 different field combinations were used to create CoMFA and CoMSIA models. Table 2 shows the statistical results obtained from these models. The most appropriate model should be determined based on the highest values of coefficient of determination of non-cross validation (R^2) and coefficient of determination of cross validation (Q^2), along with the lowest values of the number of the standard error of estimate (SEE), number of principal components (N) and significant F-test value.

Analysis of the statistical values presented in Table 2 revealed that the model with the highest Q^2 (0.527) and R^2 (0.900), lowest SEE value (0.195) and most significant F-value (36.080) with three principal components was the most optimal and best-fitting. This model used SEA fields (steric, electrostatic and hydrogen bond acceptors) with best coefficient of determination for prediction ($R_{\text{pred}}^2 = 0.899$). CoMFA model

was excluded based on the internal validation parameter which has a non-significant value ($Q^2 < 0.5$). Additionally, the large number of principal components ($N = 6$) obtained by PLS analysis contributes more significance to the rejecting of CoMFA model. As a result, the CoMSIA/SEA model has a remarkable level of stability and predictability as shown in Table 3.

For QSAR model's precision, it is common practice to graph the predicted values ($\text{pIC}_{50\text{pred}}$) against the observed biological activities (pIC_{50}). Accordingly, a desirable outcome is a line graph that bisects the positive quadrant of the pIC_{50} and $\text{pIC}_{50\text{pred}}$ axes. Also, it was confirmed that the correlation strength is high, as the slope of the model was almost equal to 1. Fig. 3 illustrates this correlation for both the training and test sets.

As illustrated in Fig. 3, there is a significant correlation between the observed and predicted pIC_{50} values, highlighting the robust ability of the model to predict the pIC_{50} values of new compounds specifically designed to treat breast cancer. This predictive ability must be validated based on Golbraikh and Tropsha's statistical criteria as presented in Table 4.

According to statistical criteria of Table 4, the proposed CoMSIA/SEA model can be classified as validated, provided that it is not considered as random modelling. For this, the stability of the proposed CoMSIA/SEA model was assessed using the Y-randomization test (Table 5).

Table 4

CoMSIA/SEA model statistical criteria.

Indicator of statistics	Score	Threshold	Comment
R_{pred}^2	0.899	Larger than 0.600	Validated
$R_0'^2$	0.899	Larger than 0.600	Validated
$R_0'^2$	0.899	Larger than 0.600	Validated
$ R_0'^2 - R_0'^2 $	0.000	Fewer than 0.300	Validated
$\frac{R^2 - R_0'^2}{R^2}$	0.001	Fewer than 0.100	Validated
$\frac{R^2 - R_0'^2}{R^2}$	0.001	Fewer than 0.100	Validated
K	0.998	$0.850 < K < 1.150$	Validated
K'	1.002	$0.850 < K' < 1.150$	Validated
$R_m^2 = R^2 (1 - \sqrt{(R^2 - R_0'^2)})$	0.872	Larger than 0.600	Validated
$R_m^2 = R^2 (1 - \sqrt{(R^2 - R_0'^2)})$	0.872	Larger than 0.600	Validated

$R_0'^2$: Coefficient of determination for the zero intercept in the predicted-versus-observed plot.

$R_0'^2$: Coefficient of determination for the zero intercept in the observed -versus-predicted plot.

K: Zero intercept slope for the test set's predicted activities vs observed activities.

K': Zero intercept slope for the test set's observed activities vs predicted activities.

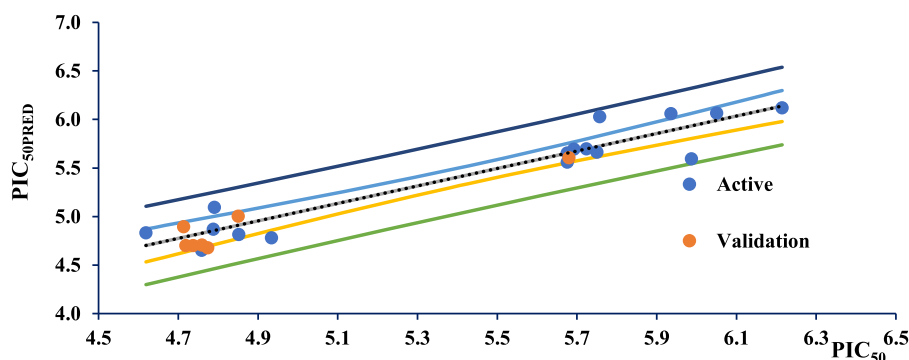


Fig. 3. Correlation between observed and predicted activity using the CoMSIA/SEA model.

Table 5
Statistics used in the Y-Randomization test.

Iterations of Y-randomization test	COMSIA/SEA model's random statistical parameters		
	Q^2_{rand}	R^2_{rand}	cR^2_p
Iteration 1	-0.202	0.791	0.603
Iteration 2	-0.214	0.726	0.525
Iteration 3	0.167	0.776	0.583
Iteration 4	-0.215	0.713	0.511
Iteration 5	0.074	0.731	0.530

The results listed in Table 5 validate and demonstrate the reliability of the model. In fact, the values of Q^2_{rand} , R^2_{rand} and cR^2_p indicate that the built model is not the result of a random correlation. Therefore, it can be used to predict the effects of potential new potential treatments for breast cancer within a certain applicability domain.

3.3. Applicability domain (AD)

The AD of the CoMSIA/SEA model was assessed using William's plot (Fig. 4), which evaluates the leverage of each compound in the training and test sets. The results indicate that the leverage value of each compound is below the warning leverage ($h^* = 0.750$), which means that all compounds fall within the AD of the model. Accordingly, all predicted

activity values for compounds are reliable and can be analyzed further.

3.4. Visualization of the best model (CoMSIA/SEA)

The best 3D-QSAR model was represented using the CoMSIA/SEA contour maps, with the most active chemical (14) as the reference molecule. Fig. 5 illustrates the steric (a), electrostatic (b), and hydrogen bond acceptor (c) fields. Blue and red contours indicate the electrostatic field, with blue outlines designating areas favoring positive electrostatics and red contours favoring negative electrostatics. Yellow and white contours represent the CoMSIA steric field, with yellow contours denoting regions that favor steric characteristics and white contours denoting regions that prefer the least steric qualities. Magenta and cyan contours represent the hydrogen-bond acceptor field, with magenta showing locations that prefer hydrogen-bond acceptor characteristics and cyan suggesting those that dislike them. These maps offer important details regarding the interactions between molecules in molecular systems.

The examination of CoMSIA electrostatic field contour maps revealed notable similarities with the corresponding CoMFA contour maps. The presence of a blue outline surrounding the Cl atom and R location suggests that electron-withdrawing groups may enhance activity, indicating that strongly electronegative atoms or groups can potentially boost the desired effect. Conversely, the red contour encompassing the hydrogen of the benzene group implies that

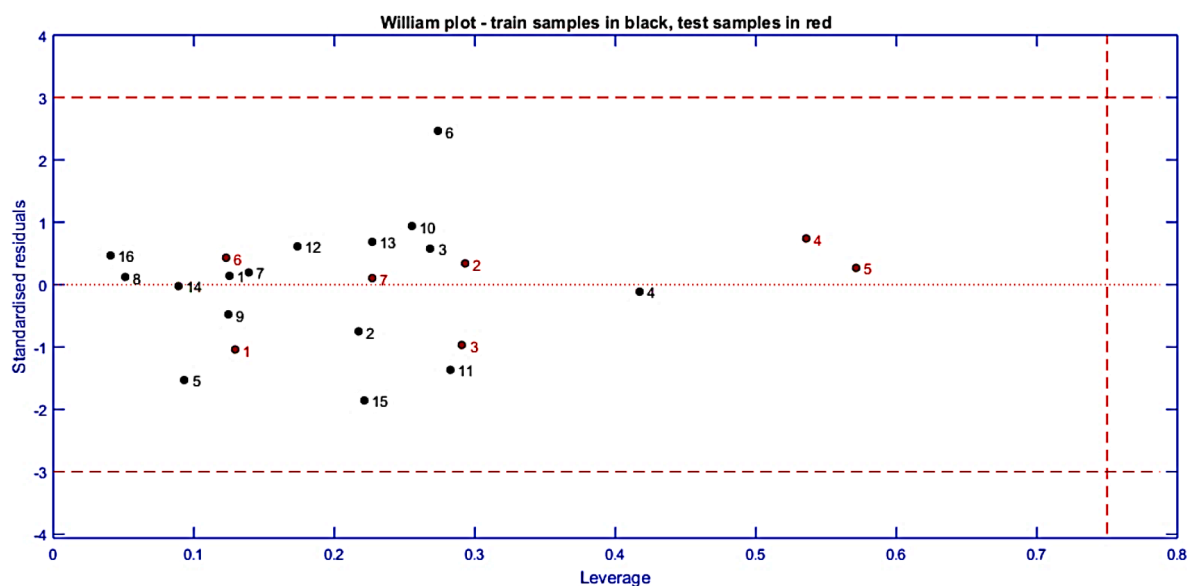


Fig. 4. CoMSIA/SEA model's applicability domain.

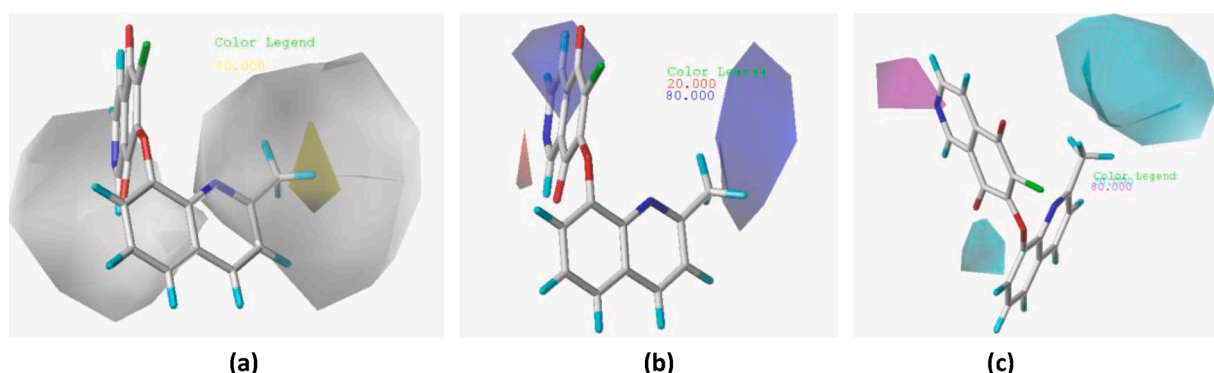


Fig. 5. Contour maps (CoMSIA/SEA) of compound 14 with 2 Å grid spacing; Steric fields (a), electrostatic fields (b) and H-bond acceptor fields (c).

Table 6
CoMSIA analysis of Fields' portion.

COMSIA/ SEA Fields	Fraction
STERIC (S)	0.072
ELECTROSTATIC (E)	0.410
ACCEPTOR (A)	0.519

substituting electron-donating groups for the hydrogen could lead to increased activity, signifying that the presence of electropositive groups or atoms in this area may enhance the anti-breast cancer properties. The yellow contours surrounding the methyl group (R) indicate that introducing bulky groups at this site may reduce activity, whereas the white contours around the benzene positions suggest a preference for less bulky substitutes. The magenta contours encircling the heterocyclic aromatic, particularly the nitrogen group, imply that the incorporation of a hydrogen bond acceptor at this site may improve activity. Similarly, the cyan contours in proximity to the methyl group (R) suggest that the

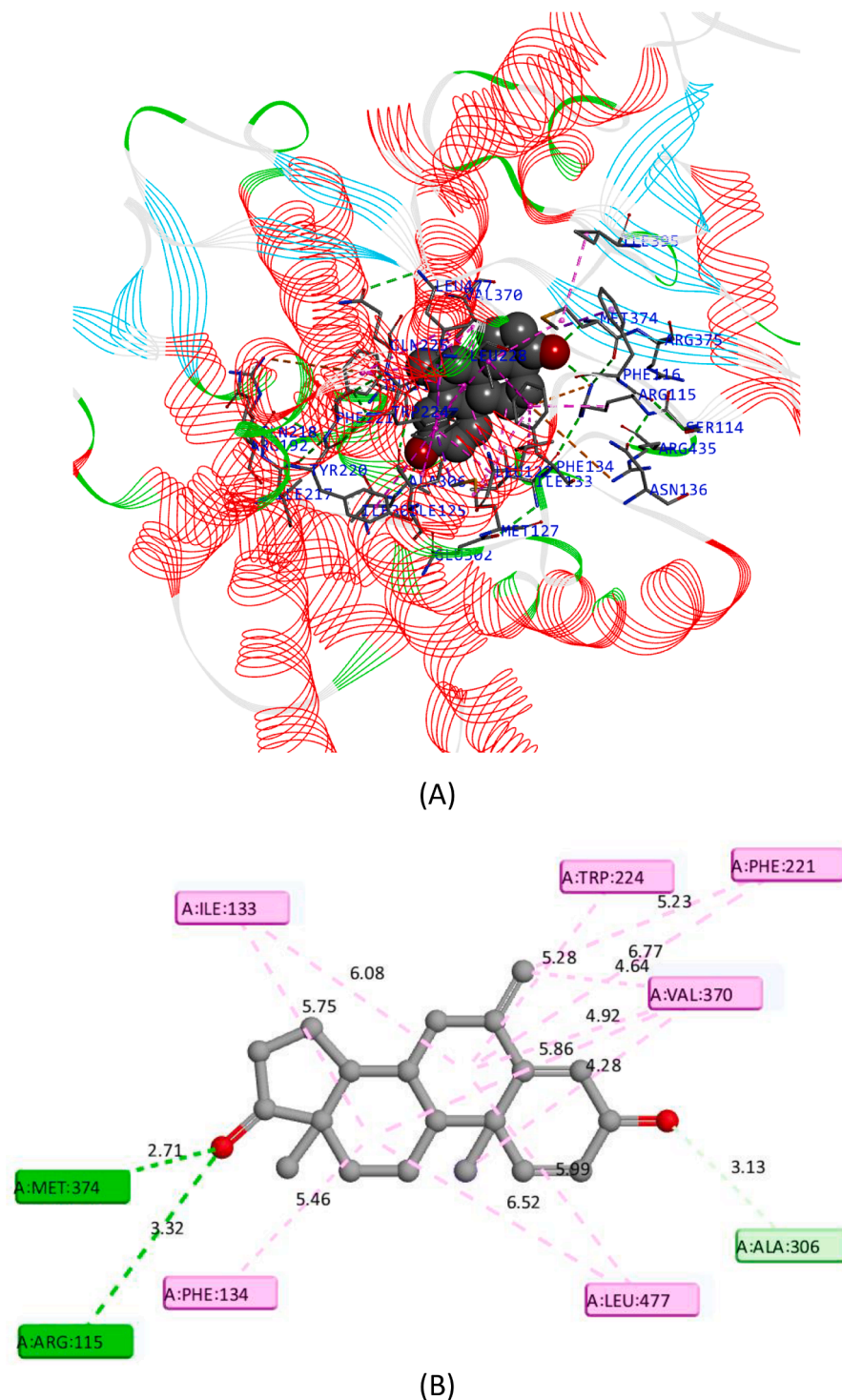


Fig. 6. 3D (A) and 2D (B) interactions Complex 3S7S-Exemestane.

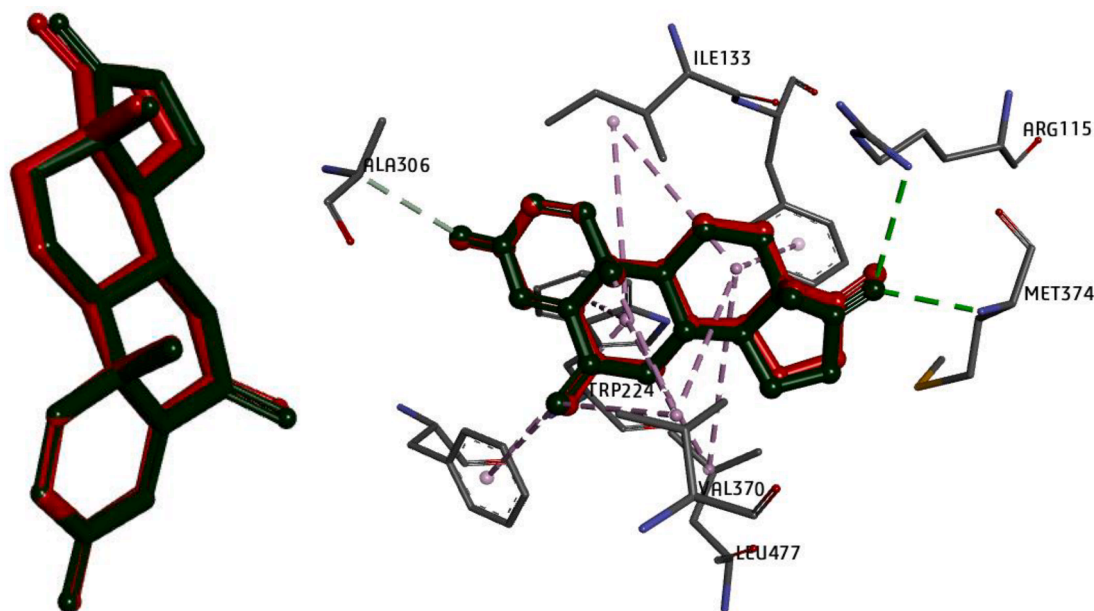


Fig. 7. Superposition between original (green color) and re-docked (red color) ligands.

addition of a hydrogen bond donor group at these positions might enhance breast cancer inhibition. These observations are elucidated by the fact that the most potent compound (14) possesses a nitrogen atom ($Y = N$) in the aromatic group, serving as a hydrogen bond acceptor, and CH_3 at the R position. In contrast, the least effective compound (22) features an aromatic group ($Y = CH$) and pyrrolidine at the R position, which is larger and bulkier than a methyl group due to the presence of the ring.

According to the CoMSIA field fraction significance values in Table 6, effectiveness against breast cancer may be successfully increased by boosting hydrogen bonding, steric, and electrostatic interactions. It could be essential to replace the R group of the most active molecule with a less steric and electronegative group, such as the $-NH_2$ substituent, that contains more hydrogen bond donors than acceptors.

3.5. Molecular docking study

In order to corroborate the results of the CoMSIA/SEA analysis and better understand the binding mechanisms of the compounds studied with the aromatase enzyme, a molecular docking study was performed. The first step aimed to elucidate the interactions between the original ligand (Exemestane) and the target protein (3S7S) via visualization of active site using the discovery studio software, as shown in Fig. 6. The predicted active site of the target protein (PDB ID: 3S7S) was found to have several key amino acids, such as MET374, ARG115, PHE134, ILE133, ALA306, LEU477, PHE221, VAL370, TRP224, and VAL373, which are very important for the molecular docking evaluation [50,51].

Exemestane, the original aromatase inhibitor, sufficient funds available hydrogen bond acceptor interactions with MET 374/ARG 115, which are essential for suppressing breast cancer, according to the identification of the protein-ligand complex interactions. We performed a re-docking of the co-ligand at the same active site (PDB ID: 3S7S), creating grid maps with 70 in all directions (X, Y, Z) and default grid space sizes of 0.375 Å in order to confirm the accuracy of the molecular docking method employed in this work. The center grid box's coordinates were determined to be 86.031 Å, 54.004 Å, and 46.404 Å, using the initial ligand position. A 3D visualization of the binding interactions between the docked ligands and the protein was done using the Discovery Studio software once docking was finished as shown in Fig. 7.

To verify the outcomes of molecular docking, we compared the

lowest energy conformation obtained from re-docking with the original ligand. The RMSD of the superimposed structures of the two ligands was 0.251 Å, which confirms the accuracy of the docking method. After confirming the active site, we selected the most active (compound 14) and the least active (compound 22) compounds for docking to represent the various types of binding with the aromatase enzyme. The interactions and affinity energies between the aromatase (PDB ID: 3S7S) and these compounds are illustrated in Fig. 8.

Fig. 8 reveals that specific compounds exhibit robust hydrogen bonding and pi-interactions that encompass both electrostatic and steric interactions with critical amino acids in the protein. The most active molecule formed two crucial hydrogen bond acceptors between nitrogen and Met 374 (2.300 Å), the carbonyl group with ARG 115 (2.830 Å), and pi-donor hydrogen bonds with VAL 373. ARG 115 and MET 311 also engaged in electrostatic interactions with the molecule via pi-cation bonding, while CYS 437 displayed a pi-sulfur interaction. The heterocyclic rings and benzene bond with other essential amino acids through pi-interactions (Steric) as shown in Fig. 8(A). On the other hand, the least active molecule docked with CYS 437 and MET 303 through pi-sulfur interaction. Other interactions with important amino acids included pi-alkyl, pi-pi, pi-sigma, and alkyl interactions, as shown in Fig. 8(B). The analysis highlighted the possibility of improving drug efficacy by reducing binding energy and increasing the ratio of hydrogen bond donors / acceptors engaged with the receptor. Due to the different binding energies (−8.100 Kcal/mol for the less active molecule (22) and −9.600 Kcal/mol for the active molecule (14)), the active molecule exhibited a higher experimental pIC50 value. Therefore, improving the affinities of hydrogen bond and electrostatic interactions may create brand-new anti-breast cancer substances. These results support and enhance the present CoMSIA results.

3.6. Discovery of a potential new candidates for breast cancer drugs

3.6.1. Design and selection of lead compounds

In order to improve the current study compounds biological activity as a potential breast cancer drug, five compounds (Ligands 1, 2, 3, 4, and Ligand 5 (Fig. 9)) based on the visualization fields of the CoMSIA/SEA model and the interactions types shown in molecular docking study. For this, the active molecule 14 was used as a reference to model compounds (Ligands 1, 2, 3, 4, and Ligand 5) by replacing R rationally based on the fragments of Zinc database [52,53]. The newly designed ligands

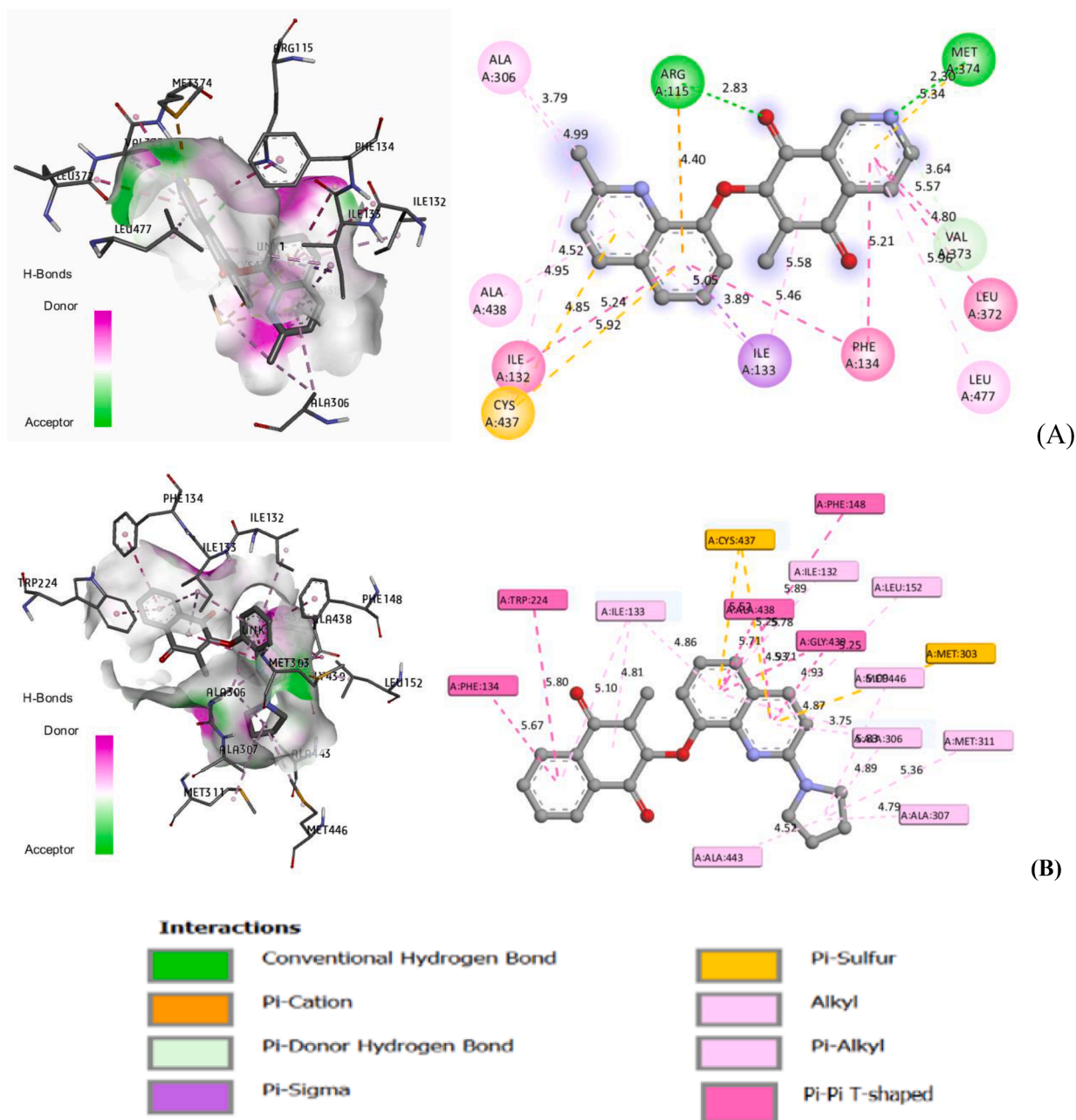


Fig. 8. Interactions of the most active (A) and least active (B) molecules into the enzyme's binding site in 2D and 3D.

underwent optimization and were aligned with a database using molecule 14 as a reference template due to its high activity level.

Results indicated that these new drug-candidates showed improved biological activities compared to molecule 14 (Table 7). It is important to note that derivatives outside the AD are not considered potential drug-candidates. Moreover, for the assessment of drug-likeness, we applied the Lipinski Rule of Five and evaluated synthetic accessibility (SA) by utilizing the online tool SwissADME. These steps in the pharmaceutical industry are crucial for selecting viable drug-candidates [43]. Table 7 summarizes the biological activities and drug-like properties expected of the designed ligands as well as CoMSIA fields and h_i values.

In Table 7, the Lipinski rule of five proposed ligands (1–5) is respected. Each of the five compounds proposed has a synthetic accessibility value ranging from 3.380 to 3.960, which falls within the range of 1 to 10. This suggests that all the developed compounds were considered potential drug-candidates due to their ease of synthesis. But,

according to leverage values h_i that must be less than h^* ($h^* = 0.750$), two molecules (ligand 3 and ligand 4) are outside of AD, and three molecules (ligand 1, ligand 2, ligand 5) were selected as drug-candidates. Furthermore, ligands 2 and 5 exhibited superior anti-breast cancer activity compared to molecule 14, which was the most potent compound in the examined series.

3.6.2. Docking investigation of newly chosen compounds

To improve the accuracy of docking interpretation, a validated 3D grid, previously employed for docking validation, was utilized to dock all the newly chosen ligands onto the protein's active site. Table 8 and Fig. 10 present the interactions and binding energies of the most potent ligands (1, 2, and 5) and the reference molecule 14 with the aromatase target.

In this study, it was observed that the binding energy of the reference molecule was higher than that of the proposed candidates, with values of -9.400 Kcal/mol, -9.700 Kcal/mol and -10.400 Kcal/mol for ligands

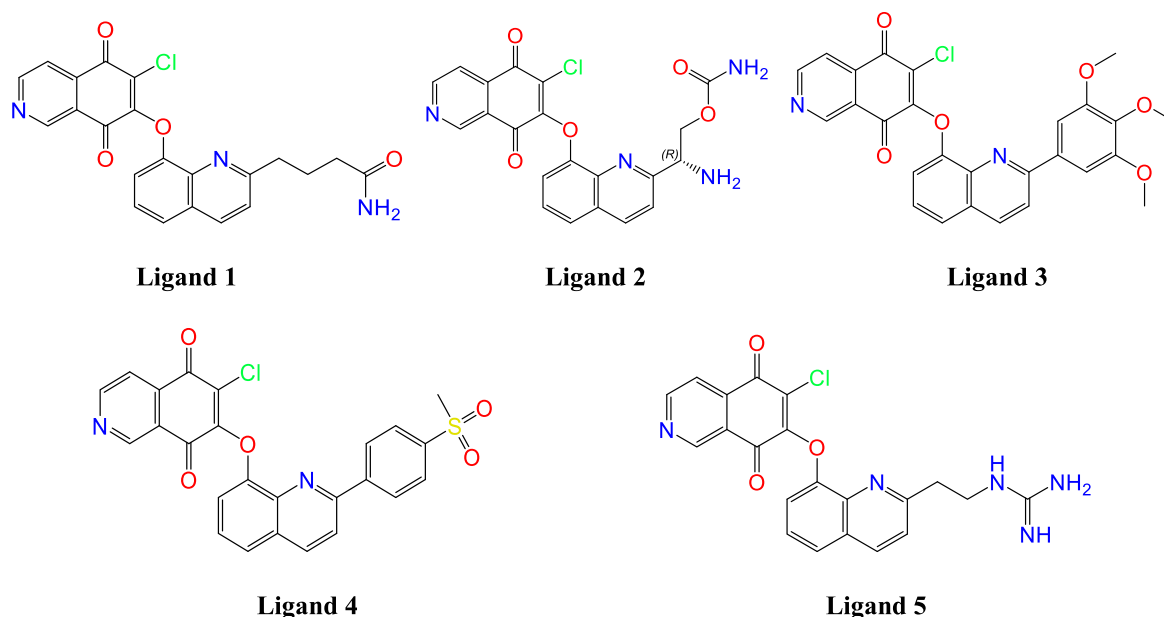


Fig. 9. Newly developed compounds' chemical structures.

Table 7

Values of COMSIA descriptors, predicted pIC_{50} , h_i , Lipinski validation and SA of the newly designed drug-candidates.

Name	ACC	Str	ELEC	pIC_{50pred}	h_i	Comments on the AD	Lipinski validation		SA
Ligand1	6.485	9.779	1.764	6.231	0.138	Inside (h_1 less than h^*)	Yes	3.380	
Ligand2	6.446	9.970	1.772	6.245	0.116	Inside (h_2 less than h^*)	Yes	3.960	
Ligand3	5.022	10.611	1.767	6.111	2.065	Outside (h_3 more than h^*)	Yes	3.840	
Ligand4	6.773	9.921	1.664	6.092	1.054	Outside (h_4 more than h^*)	Yes	3.580	
Ligand5	6.395	9.729	1.818	6.700	0.415	Inside (h_5 less than h^*)	Yes	3.440	

Table 8

Docking findings of the proposed ligands and the reference molecule 14.

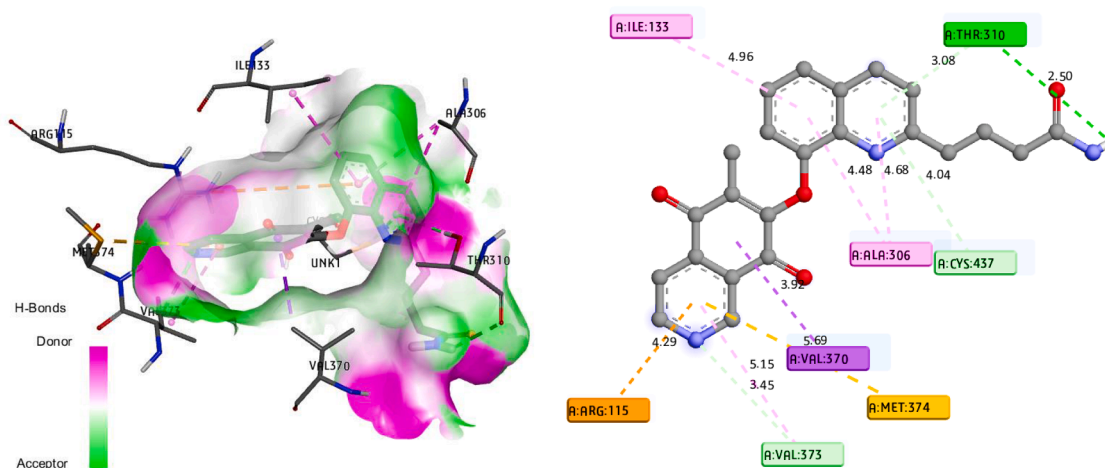
Selected ligands	Complex (Ligand - Protein)	Energy of Binding (Kcal /mol)	Number of Hydrogen-Binding (HB) interactions		Number of Pi-interactions Pi-Pi and Hydrophobic/ Steric	Number of electrostatic Interactions Pi-ions and Pi-Sulfur
			HB Acceptors	HB Donors		
Ligand 1	Ligand 1 - 3S7S	-9.400	1	3	5	2
Ligand 2	Ligand 2 - 3S7S	-9.700	1	3	6	4
Ligand 5	Ligand 5 - 3S7S	-10.400	2	1	12	8
Molecule 14	Molecule14 - 3S7S	-9.600	2	1	13	4

1, 2 and 5, respectively. This suggests that the comparatively higher pIC_{50} values of ligand 1 (6.231), ligand 2 (6.245) and ligand 5 (6.700) compared to the reference molecule (6.215) may be attributed to the stability of the ligand-complex with the lowest energy. The observed phenomenon can be attributed to the types and number of interactions between these ligands and the active site of the receptor. The strength of the binding energy and the biological activity of the ligand-receptor complex can be directly related to the number of hydrogen bond interactions involving crucial amino acids responsible for breast cancer inhibition. It's worth noting that the designed ligand 5 exhibited significant steric interactions, resulting in a higher efficacy than molecule 14 in the fight against breast cancer. Moreover, it formed strong acceptor hydrogen bonds with essential amino acids like CYS 437 and ARG 115. Furthermore, electrostatic interactions were found to play a significant role in inhibiting breast cancer. These outcomes are consistent with those of earlier research, such as 3D-QSAR and docking analysis, as well as with the experimental findings that form the basis of this study.

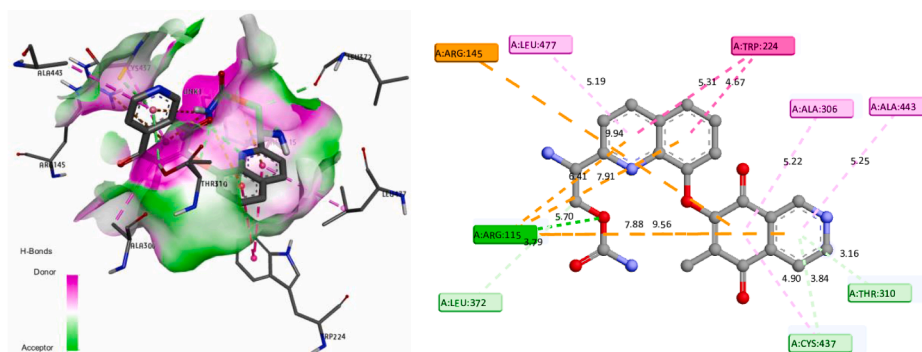
3.6.3. *In silico* ADMET prediction

To evaluate the pharmacokinetic attributes of the required compounds (ligands 1, 2, and 5), along with molecule 14 and the reference drug (Exemestane), the online tool pkCSM was employed [44,54]. The results of the ADMET features for ligand 1, ligand 2, and ligand 5 are listed in Table 9.

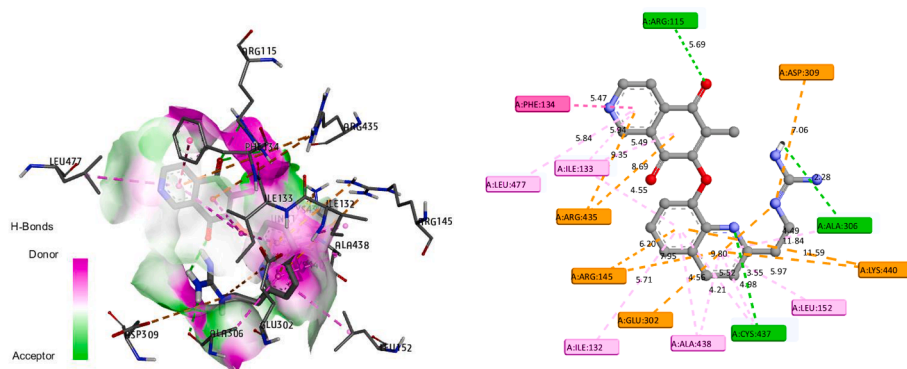
Human intestinal absorption (HIA), which refers to the uptake of substances in the human intestine, can be divided into three categories based on their percentage: weak absorption (0–20 percent), moderate absorption (20–70 percent), and great absorption (70–100 percent). The remarkable absorption of the three designed ligands in the human intestine indicates their high absorption efficiency [55]. Several studies have revealed that a VDss (volume of distribution) value greater than 0.45 indicates a significant potential for distribution [56]. Therefore, all of the designed compounds are expected to have substantial distribution potential. Regarding the typical permeability values of CNS (central nervous system) and BBB (blood-brain barrier), substances with a LogBB value below -1 are considered to have poor distribution in the brain. On the other hand, substances with a logBB value larger than 0.3 may be



Ligand 1 (Affinity = -9.400 kcal/mol)



Ligand 2 (Affinity = -9.700 kcal/mol)



Ligand 5 (Affinity = -10.400 Kcal/mol)

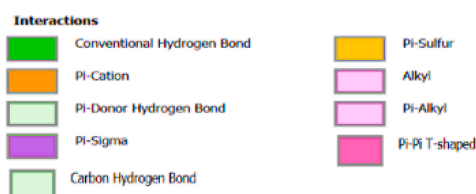


Fig. 10. 2D and 3D interaction of the newly designed ligands into the binding site of aromatase enzyme.

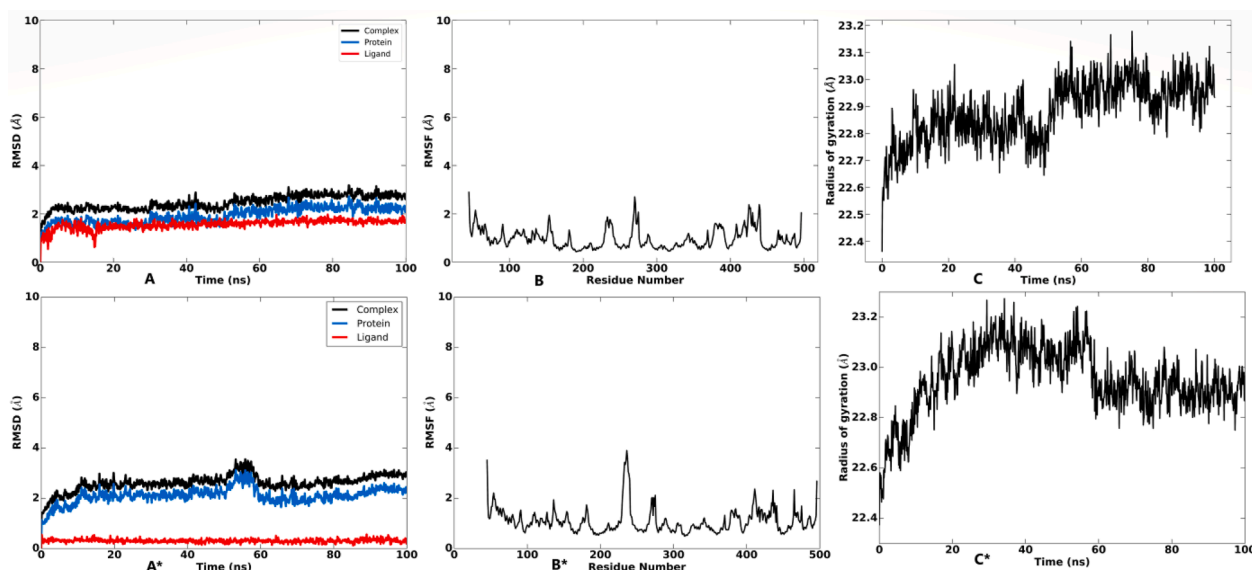
able to cross the BBB. Additionally, substances with a LogPS value greater than -2 are thought to enter the CNS. In contrast, compounds with a LogPS value less than -3 are assumed to have difficulty penetrating the CNS [57,58]. Therefore, only some of the suggested

compounds can cross these barriers. Among the enzyme families responsible for metabolizing more than 90 % of drugs during phase metabolism, CYP3A4 emerges as the most potent inhibitor [59]. Consequently, unlike the reference molecule 14, all the suggested

Table 9

ADMET characteristics of chosen compounds using the molecule 12 and exemestane as references.

	ADMET Proprieties														
	Absorption		Distribution			Metabolism							Excretion	Toxicity	
Ligands	pIC ₅₀	Intestinal absorption	VDss	BBB	CNS	CYP		CYP Inhibitor			CYP Substrate			Total clearance	AMES toxicity
						2D6	3A4	1A2	2C19	2C9	2D6	3A4	–	–	–
	–	(% Absorbed)	(Log L/kg)	Log BB	Log PS	Yes/No							log mL min ^{−1} kg ^{−1}	Yes/No	
Ligand 1	6.131	95.413	−0.253	−1.152	−3.205	No	Yes	No	Yes	Yes	No	Yes	−0.154	No	
Ligand 2 (R/S)	6.245	82.768	0.019	−1.355	−3.796	No	Yes	No	No	Yes	No	Yes	0.565	(No /Yes)	
Ligand 5	6.700	83.368	0.624	−1.672	−2.899	No	Yes	No	No	No	No	Yes	0.746	No	
Molecule 14	6	98.181	−0.249	−0.695	−1.997	No	Yes	Yes	Yes	Yes	No	No	0.030	Yes	
	0.215														
Exemestane	–	100.000	0.472	0.142	−2.267	No	Yes	No	Yes	No	No	No	1.015	No	

**Fig. 11.** (A, A*) COMPLEX RMSD, (B, B*) RMSF and (C, C*) Radius of gyration of the complexes during 100 ns MD simulation (*: Exemestane).

ligands (1, 2, and 5) were identified as either inhibitors or substrates of CYP3A4. Drug clearance, which evaluates how quickly medicines leave the body at internal concentrations, is an important parameter [60]. According to the results of Table 9, none of the new compounds raises concerns regarding drug persistence. Assessing the toxicity of candidate compounds during the early stages of drug development is crucial. In this contribution study, the AMES test was employed to evaluate the toxicity of all the novel drug- candidates [61,62]. Table 9 reveals that the designed ligands (Ligands 1, Ligand 2 (R), and 5) demonstrated non-toxicity. In contrast, the most active compound (molecule 14) was toxic. Based on the analysis of ADMET properties, it can be inferred that the most active ligand investigated in this study (Ligand 5) exhibits favorable pharmacokinetic characteristics.

3.7. Analyzing the binding stability of protein-ligand interactions during MD simulations

The best complexes of protein with bound ligand (3S7S_Exemestane) and (3S7S_Ligand 5) were evaluated for their binding stability using MD calculation run for 100 ns simulations at natural room temperature conditions. Visualization of the trajectories post simulation run revealed that all ligands remain bound to the ligand binding groove of protein pocket. Calculation of RMSD, RMSF, radius of Gyration, Hydrogen Bonding, Average center of mass (COM) distance between protein and ligand and binding free energy (MMPBSA) were carried to

assess the stability of each structure.

RMSD graphs in (Fig. 11, A and A*) show the complex, protein backbone and ligand RMSD for the structure. All RMSD curves show low values and stability after 10 ns of simulation time. The average Complex and Backbone RMSD values are 2.615 Å $\pm$ 0.321 Å and 2.103 Å $\pm$ 0.314 Å, respectively.

RMSF was calculated for the protein complex based on 'C-alpha' atoms using GROMACS program. Overall, the fluctuation intensity remains below 2.0 Å for the compound except for some residues which represent a loop or turn in the protein (Fig. 11, B and B*).

The radius of gyration analysis (Fig. 11, C and C*) shows an increasing pattern between 0 ns and 35 ns and then stable values to the end of the simulation. Overall, the ROG shows small fluctuations (less than 1 Å) throughout the whole simulation, which indicates the compactness and the stability of the protein ligand system. The Rg values varied between 22.4 Å and 23.2 Å.

The total number of hydrogen bonds formed between ligand and protein during 100 ns of the simulation time are shown in (Fig. 12, A and A*). The ligand has very weak hbond interaction during most of the simulation with significant gaps. Average Center-of-Mass Distance between ligand and protein, as well as SASA, during 100 ns of the simulation time is shown in (Fig. 12, B and B*) and (Fig. 12, C and C*). The system showed very small fluctuation of COM distance, which was within 2–3 Å which indicates that the ligand stayed in its binding site.

They were using the contactFreq.tcl module in VMD, a contact

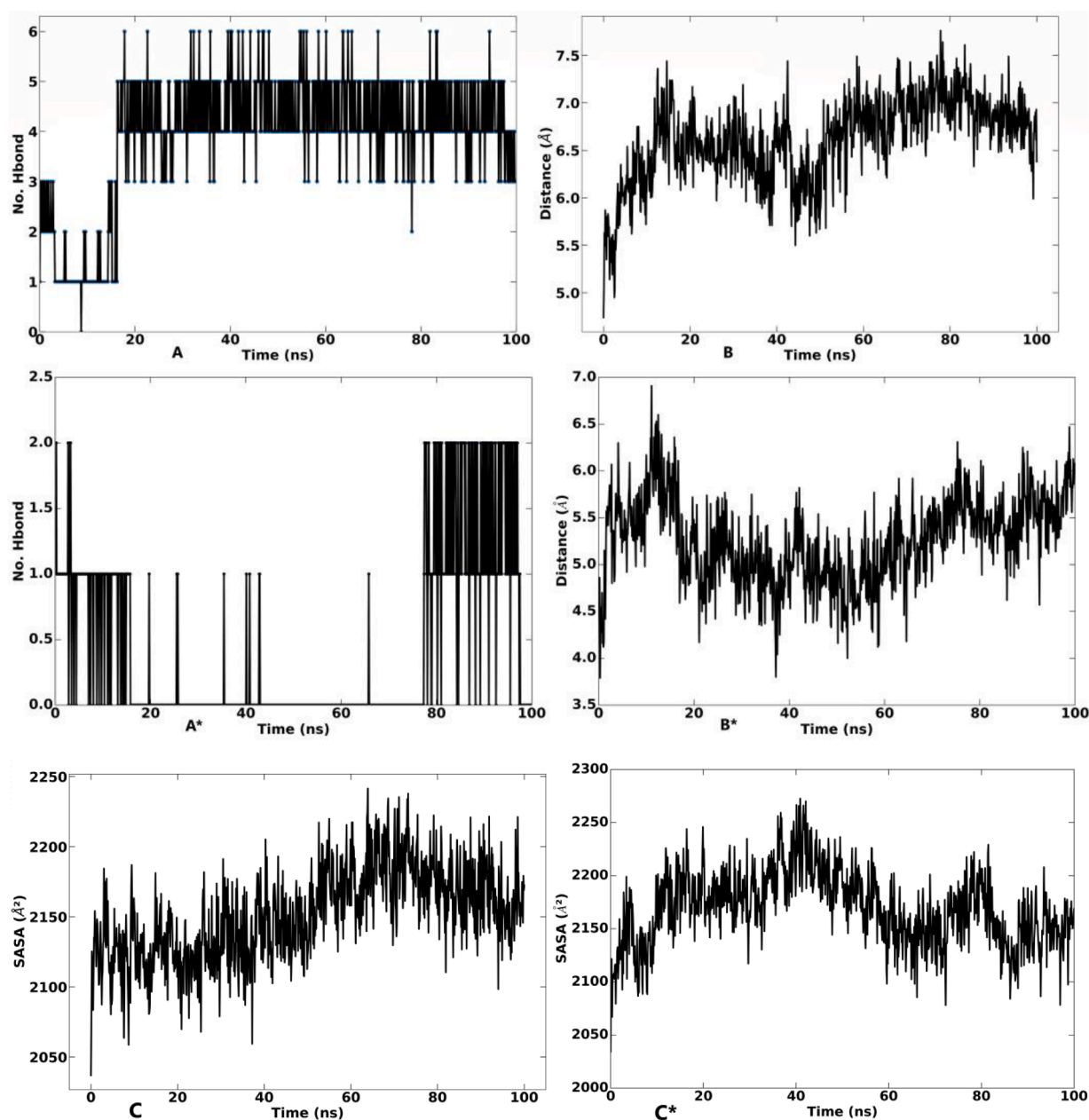


Fig. 12. (A, A*) Hydrogen Bonding, (B, B*) Average distance between Ligand and the Protein, and (C, C*) SASA (*: Exemestane).

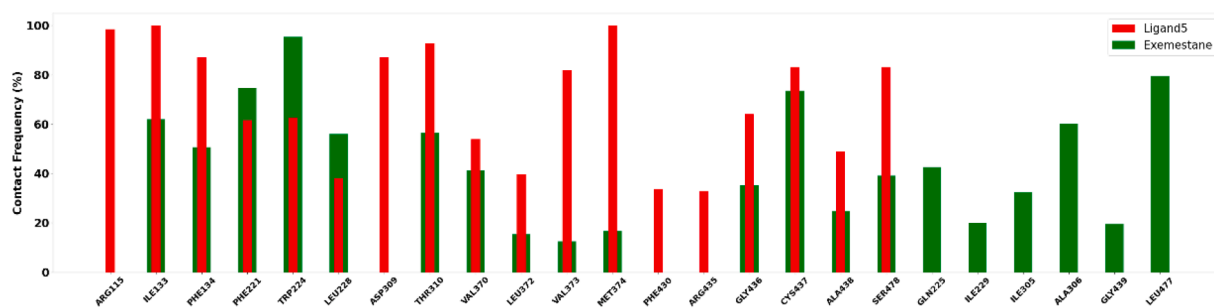


Fig. 13. Contact frequency analysis of Ligand 5 (red) and Exemestane (green).

frequency (CF) study was conducted to assess further the binding between the protein and the tested ligands. Plate_number_1, Plate_number_2, Plate_number_3, Plate_number_4, Plate_number_5,

Plate_number_6, Plate_number_7, Plate_number_8, and Plate_number_9 had the highest contact frequency throughout all simulations (Fig. 13). The Bio3D application of R was used to calculate the complexes'

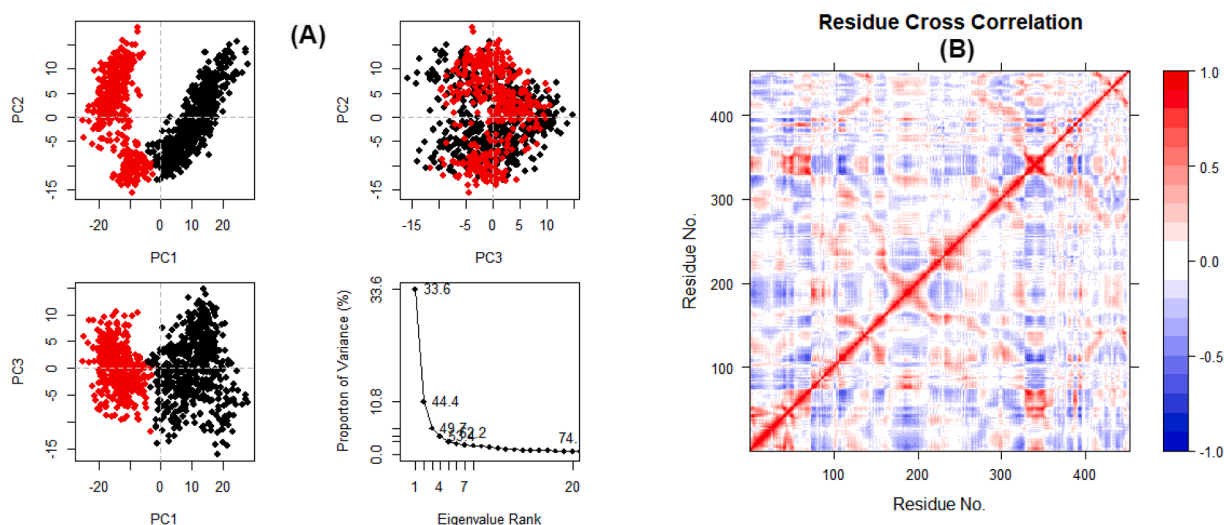


Fig. 14. Principal component analysis (A) and analysis of the dynamic cross-correlation matrix (DCCM)(B).

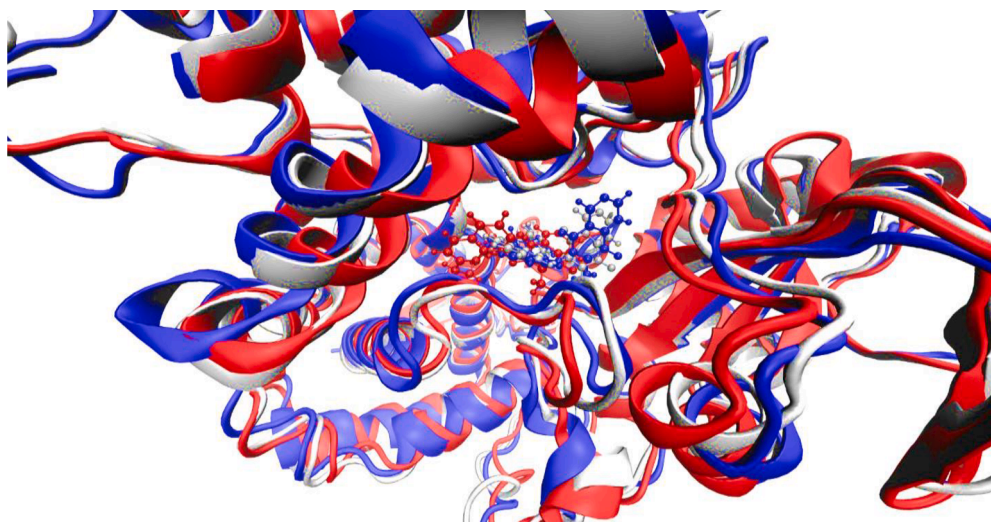


Fig. 15. The aligned structures of Ligand5-3S7S during simulation; red 0 ns, white 50 ns, blue 100 ns.

Principal Component Analysis (PCA), as shown in Fig. 14, Column A. The same algorithm was used to determine protein residues' dynamic cross-correlated movements (DCCM). As seen in Fig. 14, Column B, colors ranging from red to white to blue represent the strength of correlated motion, with blue hues denoting a negative correlation while showing no correlation and red denoting positively linked motions between residues. Fig. 15 depicts the protein-ligand complex's ligand-protein alignment during simulation time. Fig. 16 illustrates the ligand's 2D interaction diagram with the protein at the simulation's last instance (a 100 ns snapshot), demonstrating the ligand's stability. Additionally, it demonstrates various interactions between some of the residues with high CF% and the ligand. The Molecular Mechanics/Poisson Boltzman Surface Area (MM/PBSA) method was selected for rescoring complexes because it is the fastest force field-based method that computes the free energy of binding, as compared to the other computational free energy methods, such as free energy perturbation (FEP) or thermodynamic integration (TI) methods. The MM/PBSA calculation was performed using g-mmpbsa software. The calculated binding free energies are shown in Table 10 [63].

With regard to the molecular dynamics (MD) analysis of ligand positioning and protein-ligand complex stability, Ligand 5 was found to occupy the same aromatase active site responsible for breast cancer

inhibition predicted by the molecular docking study. Notably, no substantial positional changes were detected among the interacting residues before and after simulation, underscoring the stability of the Ligand 5-Aromatase complex. These results underline the importance of the ligand's ability to maintain a stable position in the protein's active site and to establish strong interactions with key residues. Furthermore, the residues involved in hydrogen bonding remained consistent throughout the initial molecular docking and the subsequent MD simulations. These results suggest that these particular residues play a vital role in the interaction with the ligand, emphasizing the compound's potential as a breast cancer inhibitor. As a result, the highly active compound (ligand 5) may be considered a promising candidate for further investigation into breast cancer inhibition via in vitro and in vivo experiments. By exploring structural modifications based on this ligand as template, research could generate a series of potentially more effective or specific derivative compounds. This rational and targeted approach represents a promising strategy for the development of new breast cancer therapies, broadening the perspectives of drug research.

4. Conclusion

In order to discover new drug-candidates, the principal goal of this

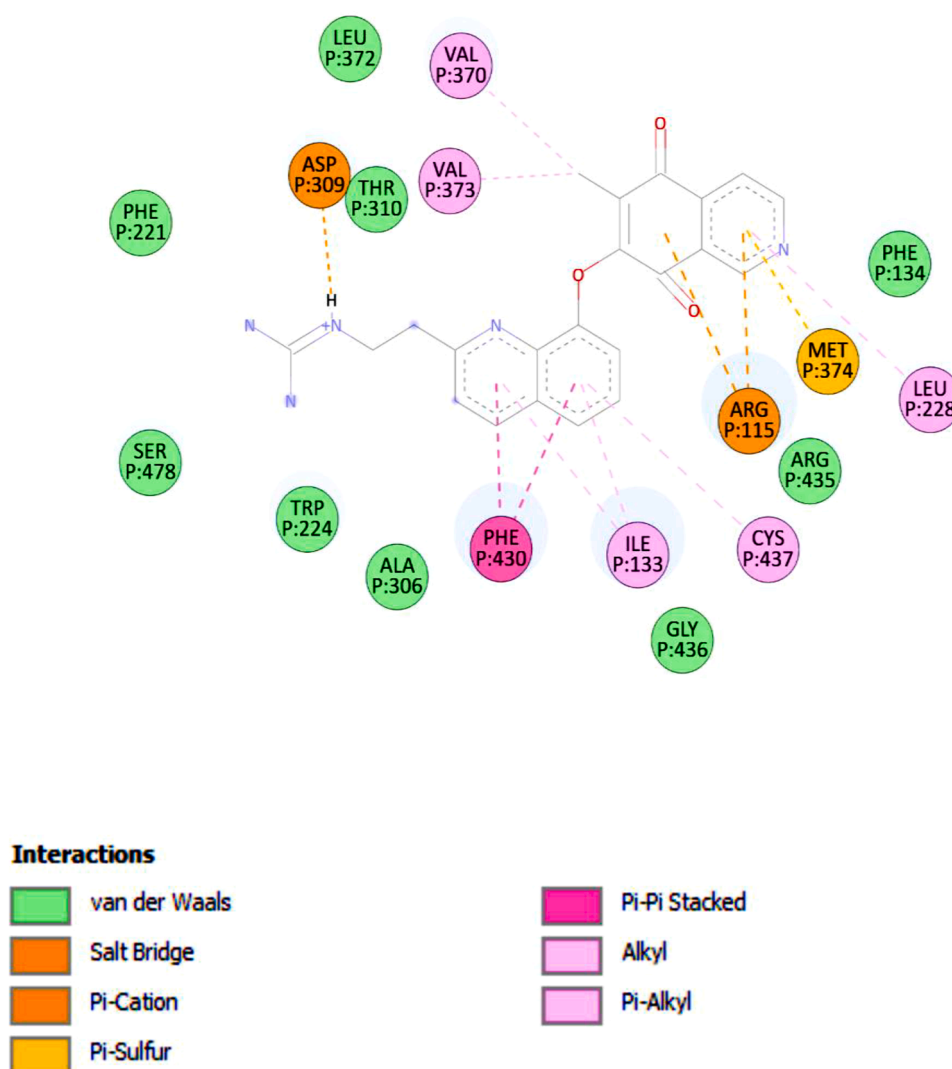


Fig. 16. A two-dimensional depiction of the interactions between Ligand 5 and the protein residues at 100 ns.

Table 10

Calculated binding free energies of tested compounds [kJ/mol].

Complex	ΔG	Van der Waal energy	Electrostatic energy	Polar solvation energy	SASA energy
3S7S_Ligand 5	-154.206 +/- 53.601	-171.208 +/- 12.197	-113.278 +/- 100.473	153.019 +/- 20.303	-22.738 +/- 0.845
3S7S_Exemestane	-132.023 +/- 22.179	-133.406 +/- 11.864	-78.462 +/- 23.651	96.561 +/- 13.790	-16.716 +/- 0.957

contribution study was to investigate the potential of quinone and quinoline derivatives as agents for the treatment of breast cancer. To achieve this, advanced computational methodologies were used, including QSAR modeling, molecular docking, in silico ADMET analysis, and molecular dynamics simulations. The accuracy of the predictive model, established by PLS regression, was validated by several validation techniques, affirming the reliability of the results. Molecular docking of the most and least active ligands was performed prior to the ADMET studies. The aim of this step was to verify the fields of the best CoMSIA/SEA model and the binding modes of the aromatase enzyme, as well as the detailed interactions of the new drug-candidates designed on the basis of this reliable model. Using molecular dynamics simulations, the study confirmed the accuracy and authenticity of the interaction between the aromatase enzyme and the best active compound (ligand 5), which was selected using a virtual screening technique (AD, SA and ADMET). Accordingly, these findings demonstrate that this ligand is highly stable upon binding to the aromatase enzyme, showing its

strength as a starting point for more drug discovery attempts to develop effective breast cancer treatments. Finally, it is envisaged that the chemical structures of ligand 5 could be modified to design new anti-cancer drugs which could be tested in vitro and in vivo.

Availability of data and materials

The data availability is included in the manuscript.

CRediT authorship contribution statement

Said El Rhabori: Conceptualization, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Marwa Alaqarbeh:** Conceptualization, Data curation, Formal analysis, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Abdellah El Aissouq:** Investigation,

Visualization. **Mohammed Bouachrine**: Validation, Visualization. **Samir Chtita**: Conceptualization, Investigation, Methodology, Project administration, Supervision, Validation, Visualization. **Fouad Khalil**: Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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